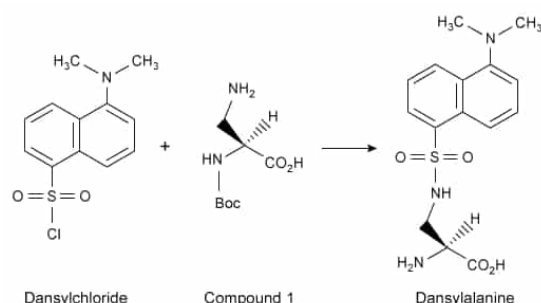


Fluorescent amino acids are useful for labeling proteins because they can be visualized spectroscopically, facilitating the study of protein structure and interactions. The noncanonical amino acid dansylalanine, a fluorescent amino acid that is sensitive to the hydrophobicity of its environment, is a good candidate to incorporate into proteins because of its small size and fluorescent capabilities. Dansyl chloride and Compound 1 were used to synthesize dansylalanine, as shown in Reaction 1. Dansylalanine absorbs maximally at 340 nm and emits at approximately 540 nm.



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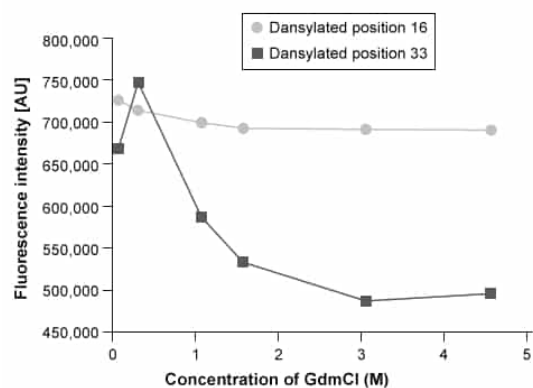


Figure 1 Plot of dansylalanine fluorescence at different concentrations of GdmCl

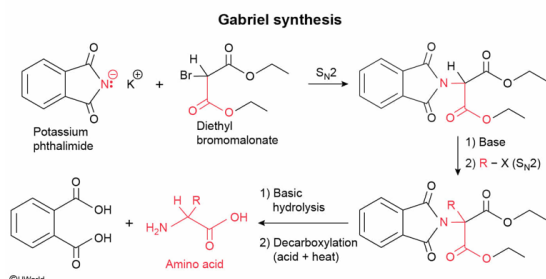
If researchers used the Gabriel synthesis to make the dansylalanine backbone, which of the following statements is true?

- ✔ ☒ A. Potassium phthalimide is a starting material. [56%]
- ☐ B. Diethyl malonate is a starting material. [10%]
- ☐ C. Potassium cyanide is a starting material. [10%]
- ☐ D. An aldehyde is a starting material. [21%]

Correct

55% answered correctly

Explanation:



The **Gabriel synthesis**, also known as the malonic ester synthesis, is a method used to make primary amines, including **α -amino acids**, without overalkylation of the amine. The amine is generated from potassium phthalimide, a "protected" form of ammonia that prevents multiple alkylations due to the steric hindrance of the phthalimide group. The synthesis begins with an S_N2 reaction, where **potassium phthalimide** is the nucleophile that attacks an alkyl halide.

The amine is then "deprotected" by removal of the phthalimide group.

For the synthesis of amino acids, potassium phthalimide is the nucleophile that attacks the electrophilic carbon of diethyl bromomalonate, and bromide is the **leaving group**. The α -carbon is deprotonated with a base and attacks an alkyl halide to add the amino acid side chain via another S_N2 reaction. The phthalimide is removed through a basic hydrolysis reaction, yielding a dicarboxylic acid. Lastly, the diacid is decarboxylated with acid and heat, and the resulting loss of CO_2 yields the final amino acid. The starting reagents for this synthesis are **planar** and do not have **chiral centers**; therefore, the amino acid product will be a mixture of L- and D-amino acids.

(Choice B) Diethyl malonate contains the $-CO_2$ group that becomes the carboxylic acid in the final product. However, it does not have a leaving group on the $-CH_2$ between the two carbonyls necessary for the S_N2 reaction with potassium phthalimide.

(Choices C and D) Potassium cyanide and an aldehyde are both used in the **Strecker synthesis** of an amino acid. Like the Gabriel synthesis, the Strecker synthesis provides a mixture of L- and D-amino acid products.

Educational objective:

The Gabriel synthesis is a method that uses potassium phthalimide and diethyl bromomalonate as starting materials to synthesize amino acids. The starting reagents of the Gabriel and Strecker syntheses are all planar; therefore, the product is a mixture of L- and D-amino acids.

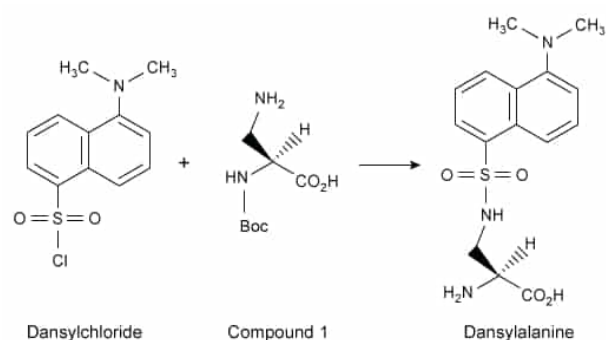
Time spent:0 sec

Subject:
Organic Chemistry

QID:401363

System:
Functional Groups and Their Reactions

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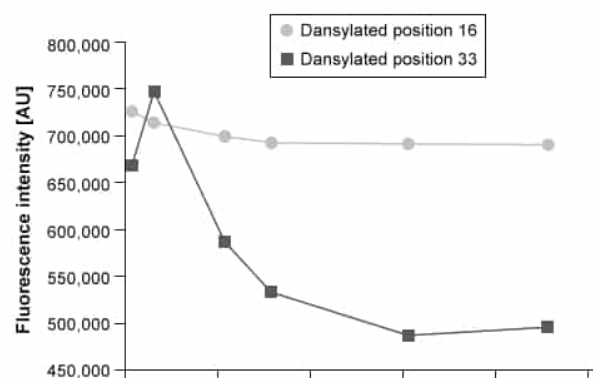




Figure 1 Plot of dansylalanine fluorescence at different concentrations of GdmCl

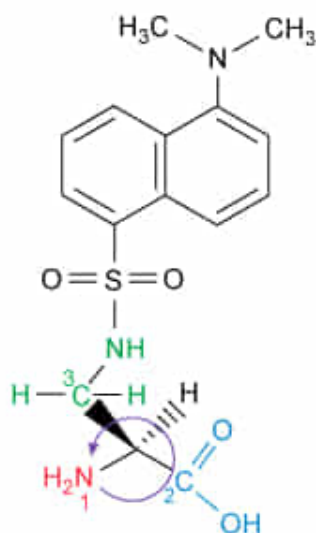
Compared to the absolute configuration of most naturally occurring amino acids, the absolute configuration of the dansylalanine shown in Reaction 1 is:

- ☐ A. different; the dansylalanine shown is an L-amino acid. [9%]
- ☐ B. similar; the dansylalanine shown is a D-amino acid. [9%]
- ☐ C. different; the dansylalanine shown has an *R* configuration. [24%]
- ✓ ☒ D. similar; the dansylalanine shown has an *S* configuration. [57%]

Correct

57% answered correctly

Explanation:



Dansylalanine

N > C, NH_2 is priority 1

C-O > C-N, COOH is priority 2

CH_2NHR is priority 3

H (priority 4) is behind the plane (dash)

Counterclockwise orientation = *S* configuration

Amino acids all have the same **general structure** consisting of a central carbon (α -carbon) bonded to an amino group, a carboxyl group, a hydrogen atom, and a variable side chain. In all canonical amino acids except **glycine**, the α -carbon is a **stereocenter**. The other amino acids can be classified by the **L/D system** or the **R/S system**. By convention, L-amino acids are drawn with hydrogen behind the plane of the backbone (dashed line) whereas D-amino acids are drawn with hydrogen in front of the plane (solid wedge).

Like **natural amino acids**, the dansylalanine shown in Reaction 1 is in the **L configuration**, with the hydrogen atom behind the plane. This configuration allows it to be incorporated into proteins in biological systems like the one described in the passage. However, L-amino acids may have either an *R* or an *S* configuration at the α -carbon. To establish the *R/S* configuration of the chiral amino acids, the **priority** of the **groups** bonded to the α -carbon must be assigned. For the natural amino acids other than glycine and cysteine, this order is:

1. -NH_2
2. $\text{-CO}_2\text{H}$
3. -R (side chain)
4. -H

The side chain of **cysteine** ($\text{-CH}_2\text{SH}$) has higher priority than the $\text{-CO}_2\text{H}$ group, so cysteine has an *R* configuration but all other natural chiral amino acids have an **S configuration**.

The side chain of dansylalanine is lower priority than the $\text{-CO}_2\text{H}$ because the side chain contains a C–N bond whereas the carbon in the $\text{-CO}_2\text{H}$ contains several C–O bonds. Therefore, the absolute configuration of dansylalanine follows the same priority order as most natural chiral amino acids and has an *S* configuration.

(Choices A and B) The dansylalanine shown is *similar* to most natural amino acids because it is in the L configuration.

(Choice C) The dansylalanine shown has an *S* configuration, not an *R* configuration.

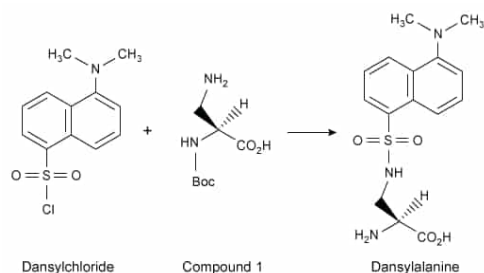
Educational objective:

All naturally occurring amino acids have an L configuration except for glycine, which is not chiral and therefore neither L nor D. Of the L-amino acids, all have an *S* configuration except cysteine.

Time spent: 0 sec
Subject:
Organic Chemistry

QID: 401364
System:
Functional Groups and Their Reactions

Fluorescent amino acids are useful for labeling proteins because they can be visualized spectroscopically, facilitating the study of protein structure and interactions. The noncanonical amino acid dansylalanine, a fluorescent amino acid that is sensitive to the hydrophobicity of its environment, is a good candidate to incorporate into proteins because of its small size and fluorescent capabilities. Dansyl chloride and Compound **1** were used to synthesize dansylalanine, as shown in Reaction 1. Dansylalanine absorbs maximally at 340 nm and emits at approximately 540 nm.



Reaction 1

A leucyl-tRNA synthetase from *Escherichia coli* was engineered to aminoacylate bacterial tRNAs containing a CUA anticodon with dansylalanine. After verifying that the tRNA/synthetase pair does not interact with endogenous amino acids or tRNAs, it was transfected into the yeast *Saccharomyces cerevisiae* along with a plasmid encoding a mutated human superoxide dismutase (hSOD). Using this system, dansylalanine was genetically incorporated into hSOD in place of either Gln-16 or Trp-33, and expressed in *S. cerevisiae*.

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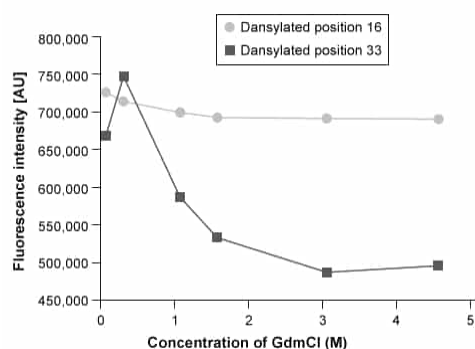


Figure 1 Plot of dansylalanine fluorescence at different concentrations of GdmCl

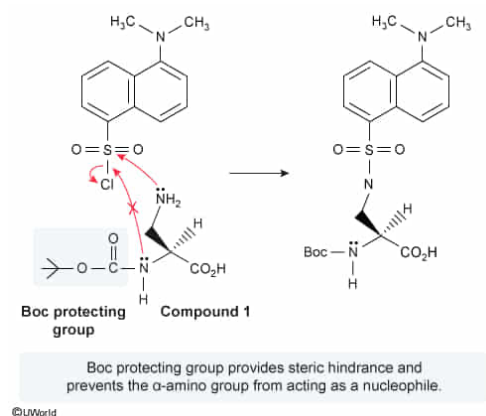
In the synthesis of dansylalanine, the Boc group on Compound **1** most likely functions to:

- ✓ ☒ A. protect the α -amino group from nucleophilic reaction. [77%]
- ☐ B. protect the C-terminus from nucleophilic reaction. [9%]
- ☐ C. make the N-terminus a better leaving group. [11%]
- ☐ D. make the carboxyl a better leaving group. [1%]

Correct

76% answered correctly

Explanation:



Organic synthesis requires reactions at specific functional groups to make the target compound.

However, some molecules have multiple functional groups that could react under the given conditions.

Protecting groups are added to organic molecules to **modify the functional group** by masking it, thereby **preventing unwanted reactions** under given reaction conditions. These groups tend to be stable to oxidation and reduction conditions and prevent nucleophilic reactions from occurring. **Particular protecting groups** exist for each type of **functional group**. After a reaction is complete, the protecting group can then be removed to reveal the original functional group.

The addition of dansyl chloride to Compound **1** requires an amine nucleophile to attack the electrophilic sulfur atom in dansyl chloride. Because two amines are found in Compound **1**, the one bonded to the α -carbon needs to be protected so it will not react. The Boc group provides steric hindrance to prevent the amino group from acting as a nucleophile.

(Choice B) Although the Boc protecting group is used to protect a functional group from reacting, the C-terminus (also called the carboxyl group) does not have a Boc protecting group attached to it.

(Choices C and D) The N-terminus and the carboxyl group of the amino acid are present in both Compound **1** and dansylalanine. Therefore, neither can participate in the reaction as a **leaving group**; instead, the chloride in dansyl chloride is the leaving group in Reaction 1.

Educational objective:

Protecting groups are organic molecules added to functional groups to mask them and keep them from reacting under a given set of reaction conditions. Different protecting groups exist for different functional groups.

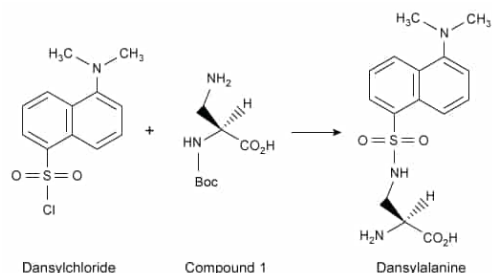
Time spent: 0 sec

Subject:
Organic Chemistry

QID: 401365

System:
Functional Groups and Their Reactions

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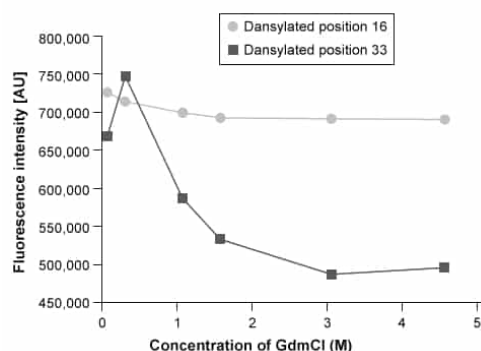


Figure 1 Plot of dansylalanine fluorescence at different concentrations of GdmCl

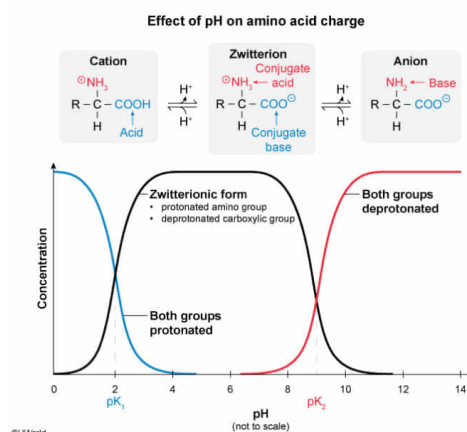
During the protein unfolding experiment, the N-terminus is:

- ☐ A. fully deprotonated and has a net charge of -2 . [5%]
- ☐ B. partially deprotonated and has a net charge of -1 . [16%]
- ☐ C. partially deprotonated and has a net charge of 0 . [25%]
- ✓ ☒ D. fully protonated and has a net charge of $+1$. [52%]

Correct

51% answered correctly

Explanation:



A protein is made up of several amino acids linked through **peptide bonds**. The free amine group at one end of the peptide and the free carboxyl group at the opposite end are known as the **N- and C-termini**, respectively. These components of the peptide can take on different charges, depending on the pH of the environment.

Under **neutral conditions** (pH ~7), the N- and C-termini are charged. Neutral pH is above the carboxyl group's pK_a of 2, causing it to become deprotonated and form its **conjugate base**, a carboxylate anion.

The **carboxyl group** in this form contains a **-1 charge**. However, neutral pH is *below* the amine group's pK_a of 9, so the amine is fully protonated and exists in the conjugate acid form (ie, an aminium ion). This **fully protonated form of the amino group** carries a **+1 charge**. The charges of the carboxylate and aminium groups cancel, and the amino acid or protein is a **zwitterion**, or a **dipolar ion**.

The passage states the unfolding experiment is done at pH 7.2. Under these nearly neutral conditions, the N-terminus is fully protonated and exists in its conjugate acid form, giving it a +1 charge.

(Choices A and B) Fully deprotonating the N-terminus would result in a -2 charge, and an N-terminus that is singly protonated would have a charge of -1, but these forms of amino acid are not observed at any pH level between 0 and 14.

(Choice C) In aqueous solutions, the N-terminus of an amino acid or protein has a net charge of 0 when the pH is greater than ~9. The unfolding experiment was done at pH 7.2

Educational objective:

In a neutral environment (pH ~7), amino acids are zwitterions (also known as dipolar ions) because the N- and C-termini are charged (+1 and -1, respectively). The carboxyl group is deprotonated and in its conjugate base form. The amine group is protonated and in its conjugate acid form.

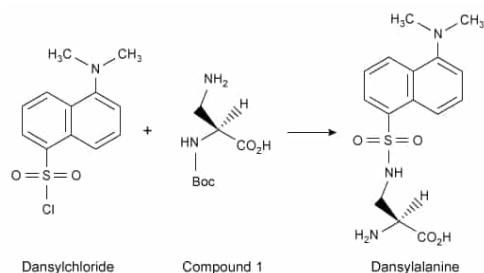
Time spent: 0 sec

Subject:
Organic Chemistry

QID: 401366

System:
Functional Groups and Their Reactions

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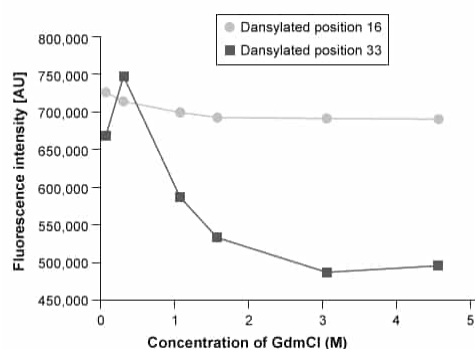


Figure 1 Plot of dansylalanine fluorescence at different concentrations of GdmCl

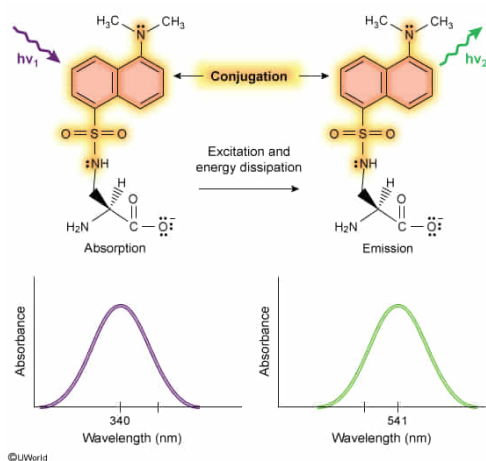
What feature of dansylalanine allows for monitoring protein unfolding?

- ☐ A. Isomerization of double bonds [8%]
- ☒ B. Conjugated pi bonds [65%]
- ☐ C. Resonance structure at the C-terminus [8%]
- ☐ D. Chirality of the amino acid [17%]

Correct

65% answered correctly

Explanation:



According to the passage, noncanonical amino acids such as dansylalanine can be used to monitor protein structure because they are fluorescent. **Fluorescence** occurs when a **photon** in the ultraviolet (UV) or visible region is **absorbed** by a fluorophore. Photons in this region have enough energy to excite certain **electrons** to **higher energy states**. After excitation, the electron loses some of its energy as heat. The remaining energy is emitted as a photon with a **longer wavelength** (ie, less energy) than the one that was absorbed.

For a molecule to absorb UV or visible light, its electrons must be sufficiently delocalized so that the energy difference between the ground and excited states corresponds to the energy of the light being absorbed. This delocalization can be achieved by a highly **conjugated** system or a system with several **alternating single and double bonds**. In general, a larger system corresponds to a higher wavelength absorbed. The **pi electrons** can be **delocalized** through the *p*-orbitals, creating an extended network of conjugated pi bonds known as **resonance** stabilization.

Dansylalanine contains an extended network of conjugated bonds through the **naphthalene** aromatic ring system and the **sulfonyl** group. This **conjugation allows fluorescence**, and therefore monitoring of protein unfolding, to occur.

(Choice A) The double bonds in the dansyl group cannot isomerize, and isomerization does not contribute to a compound's fluorescence capability.

(Choice C) Although the carboxyl group of the C-terminus can participate in resonance, the pi bond of the carboxyl group is not a part of the extended network of conjugated electrons responsible for the fluorescence of dansylalanine.

(Choice D) Dansylalanine does have a chiral center; however, chirality is not a factor that contributes to a compound's ability to fluoresce.

Educational objective:

A conjugated system of alternating single and double bonds in a molecule causes electron delocalization. This delocalization allows the molecule to absorb a photon in the ultraviolet or visible region, exciting an electron to a higher energy state.

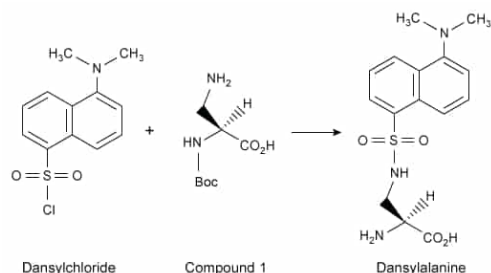
Time spent: 0 sec

Subject:
Organic Chemistry

QID: 401367

System:
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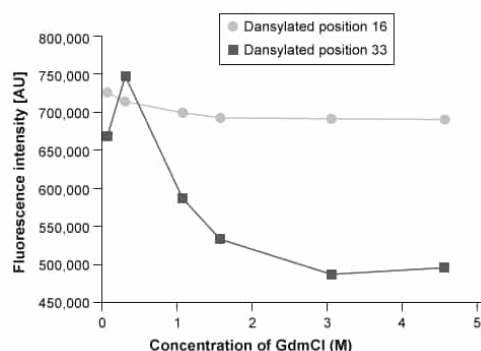


Figure 1 Plot of dansylalanine fluorescence at different concentrations of GdmCl

Of the positions where dansylalanine was incorporated, which one most likely experiences the greatest environmental change upon unfolding of the protein?

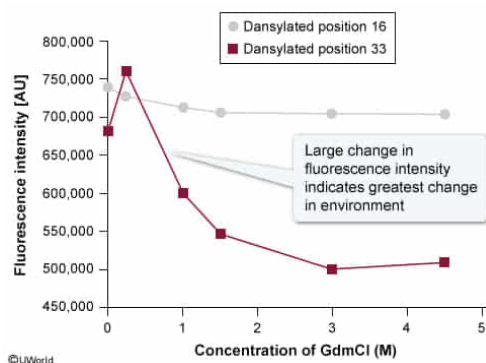
- ☐ A. Position 16, because the fluorescence intensity slightly decreases as the concentration of GdmCl increases [2%]
- ☐ B. Position 16, because it exhibits the highest fluorescence intensity at 4.5 M GdmCl [3%]
- ✔ ☒ C. Position 33, because its fluorescence intensity changes by the greatest amount as the concentration of GdmCl increases [87%]

☐ D. Position 33, because it exhibits the lowest fluorescence intensity at 3.0 M GdmCl [5%]

Correct

86% answered correctly

Explanation:



Fluorescently labeled proteins are useful for studying structure under different conditions. Many fluorophores are sensitive to environmental changes, and fluctuations in the fluorescence of these probes indicate changes in the environment around the fluorescent probe, including changes in [interactions with other residues](#). The **fluorescent amino acid** dansylalanine has a low molecular weight and can be incorporated into proteins at specified positions without significantly disturbing protein structure. This allows researchers to compare the native structure to structures in various conditions.

The passage states that dansylalanine is sensitive to the hydrophobic environment to which it is exposed. It was incorporated into hSOD at positions 16 and 33 to **study the protein's structure** as it unfolded in the presence of GdmCl. The **fluorescence** of hSOD with dansylalanine at position 33 **changes significantly** as the concentration of GdmCl increases, indicating the **environment** around dansylalanine is **changing** as the protein unfolds.

(Choice A) The fact that only a slight change in fluorescence occurs with increasing GdmCl concentration indicates that the environment around position 16 does *not* experience a significant change as the protein unfolds.

(Choices B and D) It is the *change* in fluorescence intensity as conditions change, not the *absolute value* of the intensity in any given condition, that indicates a change in local environment.

Educational objective:

Analysis of graphical data is important for drawing conclusions about what is occurring during an experiment. A change in the response of a probe indicates a change in the environment around that probe.

Time spent: 0 sec

Subject:
Organic Chemistry

QID: 401368

System:
Functional Groups and Their Reactions